



2D CrMoC₂S₆/Sc₂CO₂ multiferroic heterostructure with robust antiferromagnetic ordering, switchable electronic properties and magnetic anisotropy

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ABSTRACT

Multiferroic materials combining magnetic and ferroelectric properties are capable of integrating data storage and processing into a single device, which is promising for next generation electronics. In this work, the electronic and magnetic properties of a multiferroic heterostructure coupled with an antiferromagnetic (AFM) CrMoC₂S₆ monolayer and a ferroelectric Sc₂CO₂ monolayer, CrMoC₂S₆/Sc₂CO₂, was investigated by density functional theory method. Our results show that the CrMoC₂S₆ monolayer remains the AFM ordering under both polarization states in Sc₂CO₂ layer. Interestingly, various electronic properties can be obtained for the CrMoC₂S₆/Sc₂CO₂ heterostructure due to the interfacial charge transfer and effective electric field. In the Sc₂CO₂-P_↑ polarization state, the heterostructure is AFM quasi-half metal, and in the Sc₂CO₂-P_↓ state, the heterostructure is transitioned to be AFM semiconductor with type-II band alignments. Furthermore, the electronic property of CrMoC₂S₆/Sc₂CO₂ heterostructure can also be switched under external strains. Our study proposes a robust multiferroic candidate enabling the application for nonvolatile electrical control of 2D antiferromagnets.

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1. Introduction

Two-dimensional (2D) materials offer great opportunities for the continued miniaturization of electronics under the "More-than-Moore" trend due to its innovative structure, low energy consumption, high flexibility, high storage density, and gate-controllable operation, etc [1–3]. Among them, 2D candidates combining intrinsic magnetism and ferroelectricity play crucial roles in various electronic devices. Among which, the union of magnetism and ferroelectricity in one candidate makes the material multiferroic [4]. 2D multiferroic materials have the advantage of mutual control, i.e. charge control through magnetic fields or spin control through electric fields, which makes them ideal for "magnetic read" and "electrical write" information storage. Unfortunately, the causes of magnetism and ferroelectricity in multiferroic materials are often

opposing, i.e. the magnetism usually comes from partially filled *d* orbitals, while ferroelectricity comes from empty *d* orbitals [5,6]. Although a few intrinsic 2D multiferroic materials, such as bilayer VS₂ [7,8], Hf₂VC₂F₂ monolayer [9], VOX₂ (X = Cl, Br, I) monolayers [10–12], CuMP₂X₆ (M=Cr, V; X = S, Se) monolayers [13–15], CrBr₃ monolayer [16], etc, have been predicted, single-phase multiferroic materials are very rare due to the inherent mutual repulsion between magnetism and ferroelectricity, and the magnetoelectric couplings are very weak due to their different interaction mechanisms.

Alternatively, multiferroic *vdW* heterostructures, which integrating 2D layered ferroelectric (FE) and magnetic monolayers via the vertical stacking, have received extensive attentions [17–24], as the magnetism in magnetic monolayer can be modulated through magnetoelectric coupling by controlling the polarization direction in FE layer. Intriguing properties have been found in such multiferroic heterostructures, such as controllable magnetic axis, switchable electronic properties and magnetic configurations, etc., enabling them promising potentials for high density information storage and

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multifunctional devices. To date, a large number of 2D magnetic materials [25–31] and FE layered materials [32–37] have been experimentally synthesized or theoretically predicted, which enrich the possibility of 2D multiferroic heterostructures. In 2019, Gong et al. found that in ferromagnetic (FM) $\text{Cr}_2\text{Ge}_2\text{Te}_6$ and FE In_2Se_3 heterostructure, the magnetism of $\text{Cr}_2\text{Ge}_2\text{Te}_6$ can be switched by reversing the polarization of In_2Se_3 [38]. Recently, Guo et al. predicted that the FM and antiferromagnetic (AFM) ground states of $\text{NiI}_2/\text{In}_2\text{Se}_3$ heterostructure can be altered by changing the polarization direction [39]. Hu et al. revealed that the valley polarization of $\text{CuInP}_2\text{S}_6/\text{MnPS}_3$ heterostructure can be tuned by the inter-build electric field [40]. However, people's research progress on 2D multiferroic vdW heterostructures is still in its early stage, and the magnetic candidates used in multiferroic vdW heterostructures are usually FM monolayers. For comparison, the materials with AFM ordering are insensitive to disturbing magnetic fields and more common in reality. So, whether AFM/FE multiferroic heterostructure can effectively modulate the electronic and magnetic properties by reversing FE polarization?

Herein, we proposed a multiferroic vdW heterostructures composed of AFM CrMoC_2S_6 monolayer and FE Sc_2CO_2 monolayer. CrMoC_2S_6 is a bipolar AFM semiconductor with high Néel temperature (T_N) [41]. Sc_2CO_2 monolayer exhibits large out-of-plane electric polarization of $1.60 \mu\text{C}/\text{cm}^2$ [42] and shows interesting properties [43,44]. By changing the FE polarization direction of Sc_2CO_2 monolayer, the magnetic CrMoC_2S_6 monolayer can be tuned between quasi-half metal and semiconductor, and this switch is reversible. In addition, it is found that the magnetic axis can be controllable by changing the polarization direction. Our first-principles calculations demonstrated that charge transfer and effective electric field caused by the heterostructure interface drive the reversible electronic property of the CrMoC_2S_6 layer, from the AFM quasi-half metal to AFM semiconductor. Moreover, the switching behavior retains under biaxial tensile strains within 3%. These findings reveal the potential of multiferroic $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure in the design of future electronic and spintronic devices.

2. Computational methods

All the density functional theory (DFT) calculations were performed using the Vienna ab initio simulation package (VASP) [45,46]. The ion-electron interactions were treated with the projected augmented wave (PAW) method [47], and the general gradient approximation (GGA) parametrized by Perdew, Burke, and Ernzerhof (PBE) [48] was used as the exchange-correlation functional. The valence configurations of C, O, S, Sc, Cr and Mo atoms are $2s^22p^2$, $2s^22p^4$, $3s^23p^4$, $3d^14s^2$, $3d^54s^1$ and $4d^55s^1$, respectively. The energy cutoff was set as 400 eV. The energy and force convergence thresholds for the iteration in the self-consistent field (SCF) were set to 10^{-5} eV and 0.01 eV/Å, respectively. A k -point mesh of $9 \times 9 \times 1$ was used for structure optimization, and denser k -mesh of $25 \times 25 \times 1$ was used for electronic structure calculations. The vacuum layer of about 20 Å along the c -direction was adopted to avoid the interaction between two periodic units. DFT-D2 method [49] is used to treat the vdW weak interaction. The Hubbard U correction of $U_{\text{eff}} = 4.0$ eV was considered to describe the Coulomb and exchange interaction of Sc/Cr/Mo-3d electrons according to previous studies [50–52].

3. Results and discussions

The geometries of CrMoC_2S_6 monolayer and Sc_2CO_2 monolayer are shown in Fig. 1a and 1c, respectively, the space group of which

are P312 and P3m1 with the lattice constants of 5.70 Å and 3.41 Å, respectively. Fig. 1b, d plot the band structures of freestanding CrMoC_2S_6 monolayer and Sc_2CO_2 monolayer. Clearly, Sc_2CO_2 monolayer is nonmagnetic indirect band gap semiconductor with the bandgap of 1.79 eV, while CrMoC_2S_6 monolayer is AFM indirect band gap semiconductor with band gaps of 2.09 eV and 3.10 eV in spin up and spin down channels, respectively. The local magnetic moments of Cr and Mo atoms are $3.16 \mu_B$ and $-2.69 \mu_B$, respectively.

In the construction of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure, a CrMoC_2S_6 unit cell and $a\sqrt{3} \times \sqrt{3}\text{Sc}_2\text{CO}_2$ supercell were used, where the lattice of Sc_2CO_2 monolayer was set to match that of CrMoC_2S_6 with the lattice constant mismatch of $\sim 3.41\%$. Three typical stacking configurations (type-I, type-II, type-III) were considered for the $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure, detailed structures are described and shown in Part I and Fig. S1 in Supporting Information. In the case of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ -P $\uparrow$ configuration, the energy of type-I configuration is lower than that of type-II and type-III stackings by about 0.19 eV and 0.08 eV, respectively. While in the case of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ -P $\downarrow$ configuration, the energy of type-II configuration is approximately 0.01 eV and 0.13 eV lower than that of the type-I and type-III ones, respectively. For this reason, all calculations in the following discussions are based on the type-I stacking configuration for $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ -P $\uparrow$ (Fig. 1e) and type-II stacking for $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ -P $\downarrow$ (Fig. 1f) in this work.

The interlayer distance (d) between CrMoC_2S_6 monolayer and Sc_2CO_2 (P $\uparrow$) monolayer is 2.76 Å (Fig. 1e), which is extended to 2.82 Å in the case of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ -P $\downarrow$ (Fig. 1f). To address the stability of this $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure, we calculated the binding energy based on the equation of

$$E_b = E_{\text{CrMoC}_2\text{S}_6} + E_{\text{Sc}_2\text{CO}_2} - E_{\text{HS}}, \quad (1)$$

where $E_{\text{CrMoC}_2\text{S}_6}$, $E_{\text{Sc}_2\text{CO}_2}$ and E_{HS} are the energies of isolated CrMoC_2S_6 monolayer, Sc_2CO_2 monolayer, and the $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure, respectively. The calculated binding energy for P $\uparrow$ and P $\downarrow$ polarization states are 0.49 eV and 0.35 eV, respectively. The positive values indicate the $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure is energetically favorable and experimentally feasible.

To determine the magnetic ground state of this vdW heterostructure, four magnetic configurations (FM, Néel AFM (nAFM), zigzag AFM (zAFM), stripy AFM (sAFM)) were considered, as shown in Fig. S2. Our calculations imply that both $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ -P $\uparrow$ and $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ -P $\downarrow$ have nAFM ground state with energy lower than the FM state of 0.52 eV and 0.49 eV per formula, respectively. The energy values per unit cell relative to the total energy of nAFM state are summarized in Table S1.

The electronic band structures of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure are sensitive to the polarization states of Sc_2CO_2 monolayer. Compared with freestanding monolayer, the electronic bands of CrMoC_2S_6 monolayer supported on Sc_2CO_2 -P $\uparrow$ monolayer are downshifted, while those of Sc_2CO_2 monolayer are upshifted. Contrast case is found for the heterostructure with Sc_2CO_2 -P $\downarrow$ state. The band edge alignments of the heterostructures on both polarization states can also be seen in Fig. 2e, and 2f. As a result, the $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure in the P $\uparrow$ polarization state is turned to be AFM quasi-half metal (Fig. 2a), in which the local magnetic moments of Cr and Mo atoms are 3.08 and $-2.02 \mu_B$ (Fig. 2c), respectively. As shown in the band structures, the electrons of spin up channel are quasi-semiconducting with the valence band maximum (VBM) and conducting band minimum (CBM) coming from Sc_2CO_2 monolayer, while for the electrons from spin-down channel, they are metallic with bands crossing the Fermi level. The $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure in the P $\downarrow$ polarization state is type-II AFM semiconductor with the bandgap in spin up and spin down channels around 0.38 eV and 0.94 eV, respectively (Fig. 2b). It

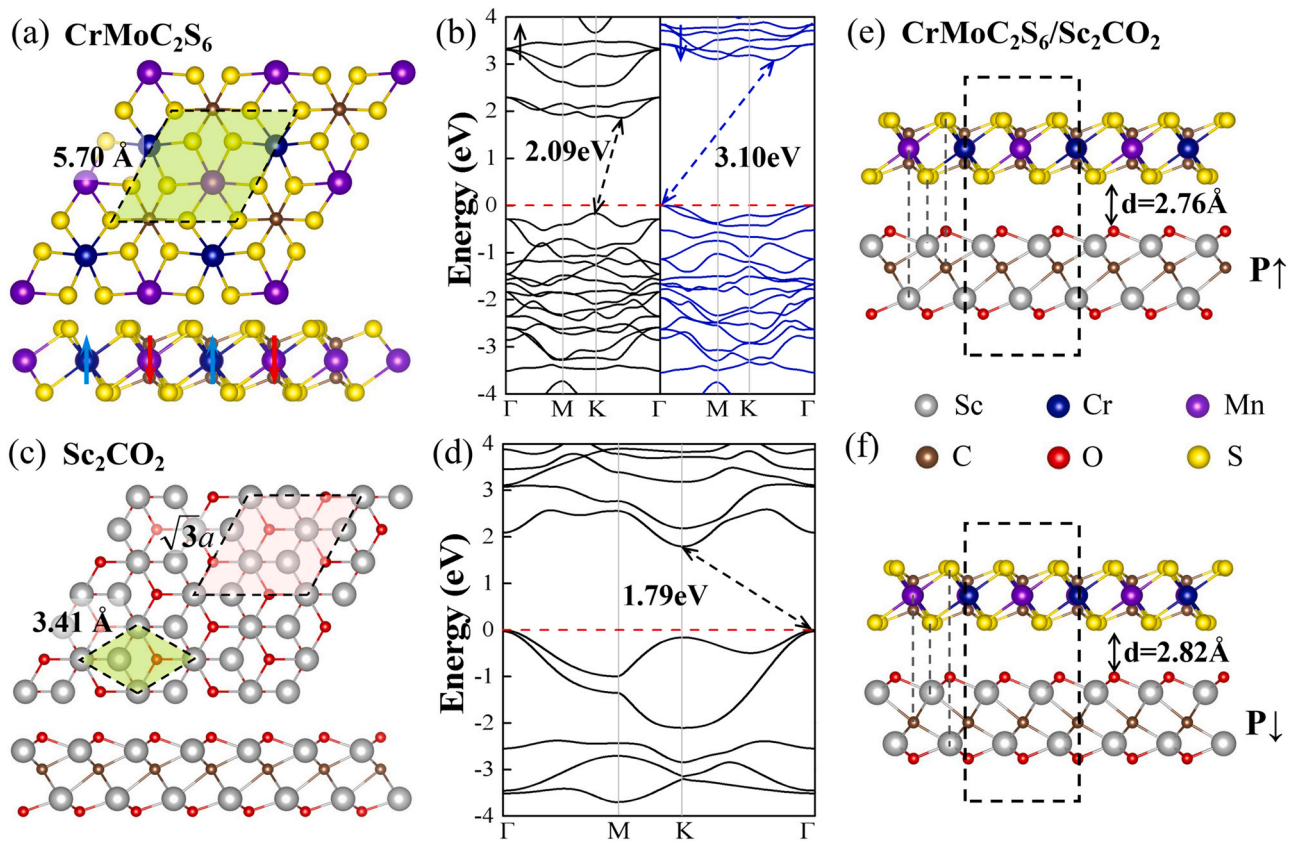


Fig. 1. Top and side views of the structures (a,c) and band structures (b,d) of CrMoC_2S_6 and Sc_2CO_2 monolayers, respectively. In the band structure plots, black and blue lines represent the bands in spin up and spin down channels, respectively. Top and side views of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ vdW heterostructures under $\text{P}\uparrow$ (e) and $\text{P}\downarrow$ (f) polarization states. The black dotted line represents the unit cell.

is displayed that the VBM mainly comes from Mo- d_{z^2} orbitals in CrMoC_2S_6 monolayer, while the CBM is mainly contributed by C/O- p_z orbitals in Sc_2CO_2 monolayer (Fig. 2d). Such switching behavior of electronic properties have also been found in $\text{FeCl}_2/\text{Sc}_2\text{CO}_2$ and $\text{Mn}_2\text{CFCl}/\text{CuBiP}_2\text{Se}_6$ heterostructures, which have respective semiconductor-metal and half metal-metal transition by reversing the FE polarization direction. [17,53].

In addition, by reversing the polarization state of Sc_2CO_2 monolayer, the easy magnetization axis of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure can be switched between the [100] and [001] directions. By considering the spin-orbit coupling (SOC) effect, we defined magnetic anisotropy energy as $\Delta E = E[001] - E[100/010]$, which is the larger energy difference between z direction and x/y direction. For $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\uparrow$ and $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\downarrow$ heterostructures, the ΔE values is 9.4 μeV and $-364.0 \mu\text{eV}$ per Cr atom, respectively, indicating the easy magnetization axis can be controllable both in-plane and out-of-plane in such 2D multiferroic heterostructure. Similar results were also found for $\text{Cr}_2\text{Ge}_2\text{Te}_6/\text{In}_2\text{Se}_3$ [38], $2\text{I-CrI}_3/\text{Sc}_2\text{CO}_2$ [21], $\text{FeBrI}/\text{In}_2\text{S}_3$ [54], $\text{Cr}_2\text{Ge}_2\text{Te}_6/\text{Sc}_2\text{CO}_2$ [19] heterostructures, in which the magnetic anisotropy energy depends on the ferroelectric polarization direction. To evaluate the spin stability of the AFM heterostructures, we performed Monte Carlo (MC) simulations to calculate the T_N of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure in $\text{P}\uparrow$ and $\text{P}\downarrow$ states. Based on 2D Heisenberg Hamiltonian model, the Hamiltonian was written as:

$$H = E_0 - \sum_{i,j} J_1 S_i S_j - \sum_{m,n} J_2 S_m S_n, \quad (2)$$

where E_0 is the energy without magnetic coupling, S_i is the spin vector in site i , J_1 and J_2 are the nearest and second-nearest exchange coupling parameters (Fig. S2a), respectively. The parameters of J_1/J_2 are obtained to be $-56.08/13.04 \text{ meV}$ and $-52.81/6.11 \text{ meV}$ for $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\uparrow$ and $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\downarrow$, respectively (see part II in Supporting Information for details). The calculated specific heat C_V as functions of the temperature is shown in Fig. 3a, and 3b, which indicates that T_N s for $\text{P}\uparrow$ and $\text{P}\downarrow$ states are about 575 K and 440 K, respectively, the T_N s higher than room temperature indicates that $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructures are highly desirable materials in electronic devices.

To gain a deep understanding of the interfacial interactions, plane-averaged electrostatic potential of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ in $\text{P}\uparrow$ and $\text{P}\downarrow$ states along the z direction were calculated. As shown in Fig. 3c and 3d, the obtained effective potential offsets are 1.83 eV and -2.07 eV for $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\uparrow$ and $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\downarrow$ heterostructures, respectively. Such opposite effective electric fields across the heterostructure lead to opposite electrons transfer behaviors in the interface, that is, about 0.066 electrons are transferred from CrMoC_2S_6 to Sc_2CO_2 monolayer under $\text{P}\uparrow$ state, while around 0.014 electrons are back transferred from Sc_2CO_2 to CrMoC_2S_6 monolayer under $\text{P}\downarrow$ state. On one hand, as shown in the band alignments in Fig. 2e, and 2f, for $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\uparrow$, the CBM of CrMoC_2S_6 monolayer in spin down channel is lower than the VBM of Sc_2CO_2 monolayer in energy compared with $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\downarrow$, which leads to larger electrons transfer between the interface. Moreover, the energy band structure of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2\text{-P}\uparrow$ becomes type-III band alignment, that is, the electrons in the valence bands of Sc_2CO_2

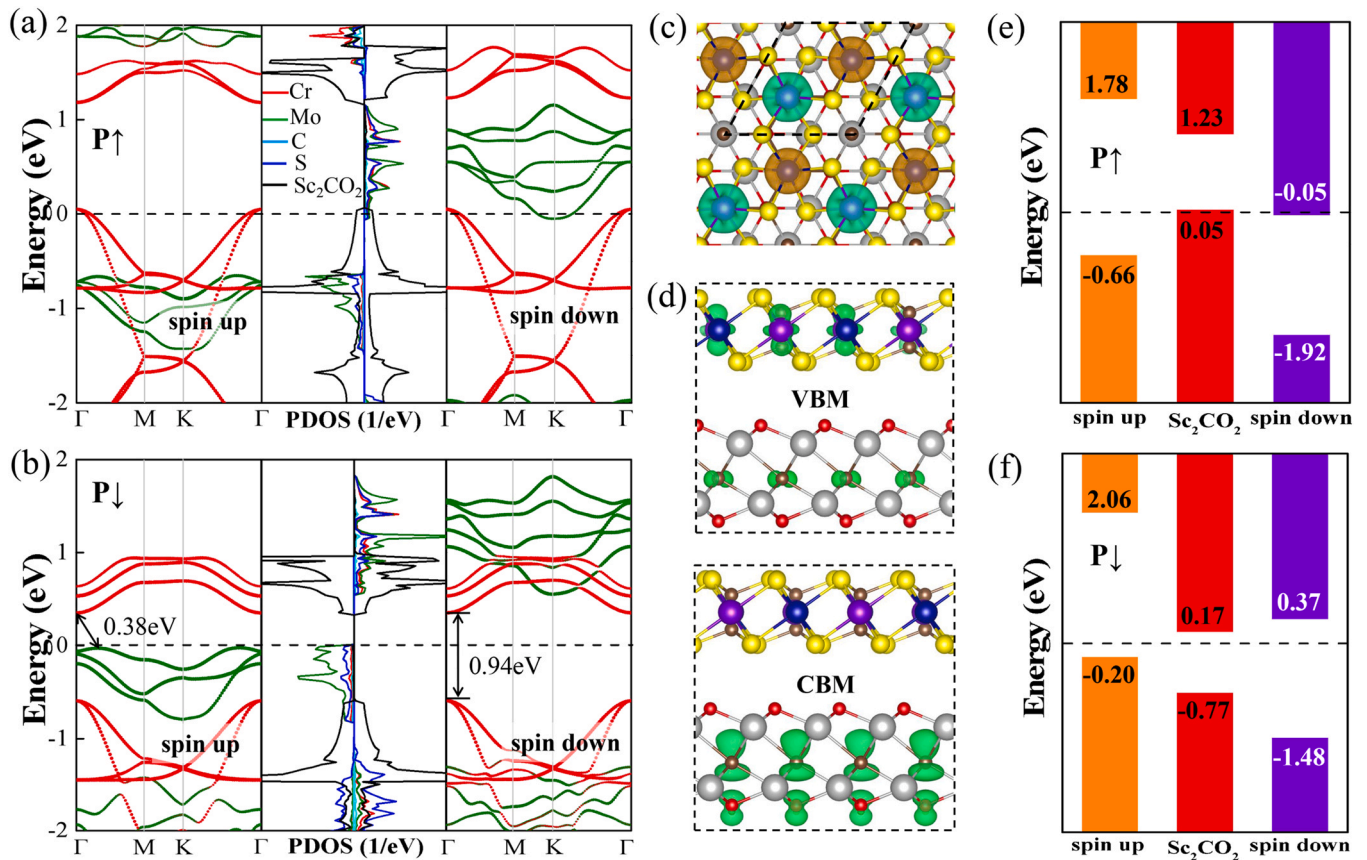


Fig. 2. Projected band structures and density of states of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ vdW heterostructures under $\text{P}\uparrow$ (a) and $\text{P}\downarrow$ (b) polarization states. In the band structures, the red and olive lines are the contributions from Sc_2CO_2 and CrMoC_2S_6 monolayers, respectively. (c) Spin density plots of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$. The orange and cyan colors represent spin up and spin down electrons, respectively. (d) VBM and CBM of semiconducting $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ - $\text{P}\downarrow$. The isosurface value is 0.02 e/Bohr^3 . (e, f) band alignments of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructures under $\text{P}\uparrow$ and the $\text{P}\downarrow$ polarization states. The orange, red and purple column stands for the band alignments of Sc_2CO_2 monolayer, the spin up and spin down band alignments of CrMoC_2S_6 monolayer.

monolayer can transfer to the conduction bands of CrMoC_2S_6 monolayer. On the other hand, the built-in electric field is from CrMoC_2S_6 monolayer to Sc_2CO_2 monolayer, which further facilitate the electrons transfer to CrMoC_2S_6 monolayer. The combined effect of the two factors results in the larger electrons transfer from Sc_2CO_2 monolayer to CrMoC_2S_6 monolayer, as shown in the charge density difference in Fig. 3e and 3f.

Furthermore, we modulated the electronic and magnetic properties of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure by applying biaxial lateral strains (ϵ), which is defined as, $\epsilon = (a - a_0)/a_0$. a and a_0 are the lattice constant under strained state and strain-free state, respectively. The positive (negative) value of ϵ represents the tensile (compressive) strain. As for the case under compressive strain, the $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure has structural deformation, therefore, it is not under consideration. Here, we considered the case of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ - $\text{P}\uparrow$ and $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ - $\text{P}\downarrow$ under $\epsilon = 1\%$ to $\epsilon = 3\%$, all the systems are AFM, the bands of CrMoC_2S_6 monolayer or Sc_2CO_2 monolayer are gradually downshifted or upshifted, respectively. Therefore, the overlap between VBM of Sc_2CO_2 monolayer and CBM of CrMoC_2S_6 monolayer are enlarged, and it turns from quasi-half-metal to metal for $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ - $\text{P}\uparrow$ (Fig. 4a-c), while the semiconducting behavior maintains for $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ - $\text{P}\downarrow$ (Fig. 4d-f). In conclusion, the electronic property of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$

heterostructure can be switched by reversing the FE polarization direction under biaxial tensile strains.

In Fig. 5a, we propose a memory model based on $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ multiferroic heterostructure for high-density data storage. First, the data writing process in this device can be controlled by switching the polarized states in FE Sc_2CO_2 monolayer. Secondly, in the data reading process of such multiferroic device, a quasi-half metallic and semiconducting state having distinct electric signals can be controlled by the polarization direction of FE Sc_2CO_2 monolayer, which can be realized by imposing external electric field, therefore, the data can be read by detecting the signal differences with lossless current. Moreover, the switching behavior is non-volatile since both Sc_2CO_2 - $\text{P}\uparrow$ and Sc_2CO_2 - $\text{P}\downarrow$ states are stable even if the electric field is removed. In addition, for the case of $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ - $\text{P}\downarrow$, the CBM and VBM of CrMoC_2S_6 monolayer are energetically higher than that of Sc_2CO_2 monolayer in spin up channel (Fig. 2f), such typical type-II band alignment will facilitate the efficient separation of photogenerated electron-hole pairs, as shown in Fig. 5b, which will result in an efficient spin-polarized photocurrent, although the electron-hole recombination can occur in Sc_2CO_2 monolayer in spin down channel as the band alignment is type-I. Such properties suggest that $\text{CrMoC}_2\text{S}_6/\text{Sc}_2\text{CO}_2$ heterostructure is a potential candidate in photoelectric devices.

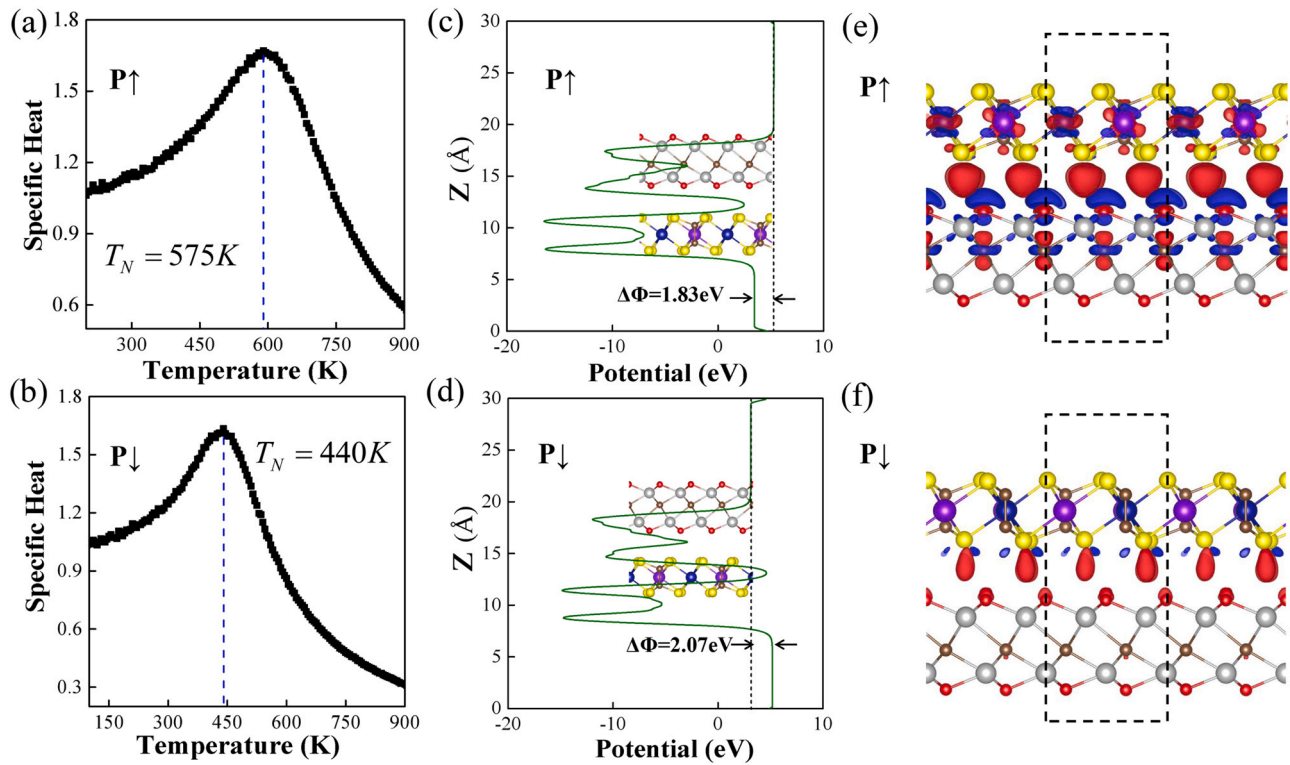


Fig. 3. Specific heat as a function of temperature for CrMoC₂S₆/Sc₂CO₂-P_↑ (a) and CrMoC₂S₆/Sc₂CO₂-P_↓ (b), respectively. (c, d) xy-plane average electrostatic potentials, and (e, f) charge density differences of CrMoC₂S₆/Sc₂CO₂ heterostructures under P_↑ and the P_↓ polarization states. The red and blue colors in (e, f) denote electron depletion and accumulation in charge density difference, respectively. The isosurface value is 0.005e/Bohr³.

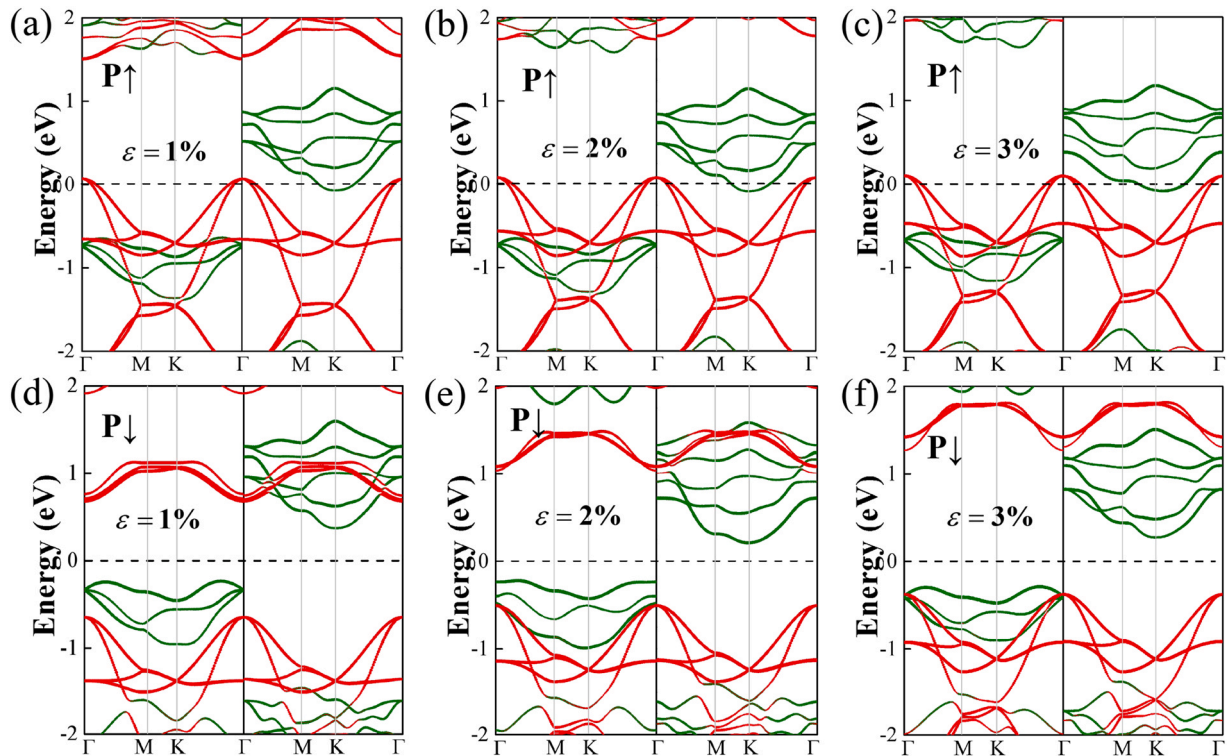


Fig. 4. Band structures of CrMoC₂S₆/Sc₂CO₂ heterostructure in P_↑ and P_↓ polarization states under biaxial tensile strains from 1% to 3%. The red and olive lines are the contributions from Sc₂CO₂ and CrMoC₂S₆ monolayers, respectively. The left and right panels denote spin up and spin down channel, respectively.

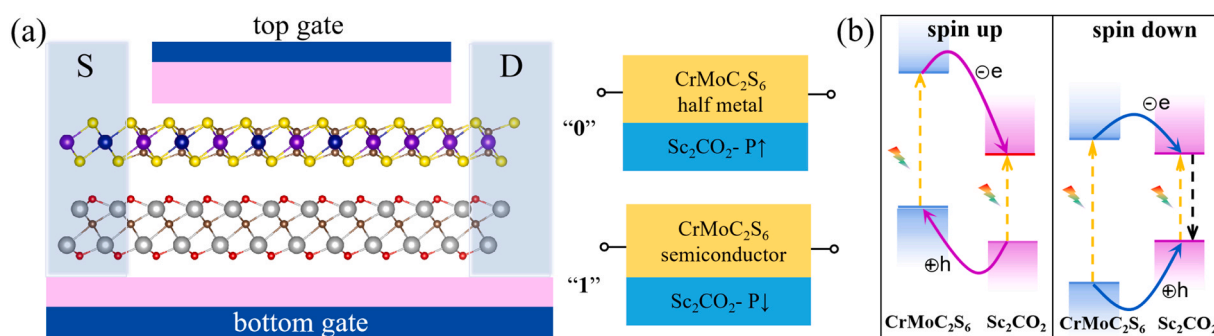


Fig. 5. (a) The schematic of multiferroic memory device based on CrMoC₂S₆/Sc₂CO₂. (b) Schematic of spin-dependent charge transfer processes in CrMoC₂S₆/Sc₂CO₂-P_↑ state under the illumination condition. Yellow and black dashed arrows represent photoexcitation process and recombination process.

4. Conclusions

In summary, we investigated the electronic properties of multiferroic CrMoC₂S₆/Sc₂CO₂ heterostructure by first principles calculations. It is revealed that the heterostructure can be switched between quasi-half metal and semiconductor by reversing the FE polarization direction of Sc₂CO₂ monolayer, which is the combined effect of band alignments and the built-in electric field. The CrMoC₂S₆/Sc₂CO₂ heterostructure has Néel temperature of 575 K and 440 K in P_↑ and P_↓ states, respectively, higher than room temperature. And the easy magnetization axis can be controllable both in-plane and out-of-plane in such multiferroic heterostructures. Moreover, the properties of CrMoC₂S₆/Sc₂CO₂ heterostructure can also be altered by switching the polarization direction under lateral biaxial tensile strains, in which CrMoC₂S₆/Sc₂CO₂-P_↑ turns from quasi-half-metal to metal. These findings suggest that the CrMoC₂S₆/Sc₂CO₂ heterostructures are potential candidates in designing the next generation of all-electrically controlled antiferromagnetic spintronic devices, new nonvolatile memory devices and photoelectric devices.

CRediT authorship contribution statement

Nini Guo: Investigation, Data analysis, Writing – original draft. **Xiangfei Zhu:** Investigation, Data analysis. **Huijie Lian:** Data analysis. **Tianxia Guo:** Investigation. **Zijin Wang:** Investigation. **Huiyang Zhang:** Data analysis. **Xiaoqing Yao:** Project administration and revise. **Bing Wang:** Revise, Software, Formal analysis. **Xiuyun Zhang:** Supervision.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2023.169962](https://doi.org/10.1016/j.jallcom.2023.169962).

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